SUPPLY CHAIN ANALYSIS USING POWER BI

Client Project Report

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Tools & Technologies used

Power BI

Power Query

DAX

MYSQL Database

Data Visualization

Project Type

Client Project

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1. INTRODUCTION

Supply Chain Management plays a vital role in ensuring the efficient movement of products from manufacturers to customers while optimizing costs, maintaining product quality, and improving customer satisfaction. Organizations generate large amounts of supply chain data related to inventory, suppliers, logistics, manufacturing, and product quality. Analyzing this data helps businesses make informed decisions and improve overall operational performance.

This project, **"Supply Chain Analysis Using Power BI,"** aims to transform raw supply chain data into meaningful business insights through interactive dashboards and visualizations. The analysis focuses on key performance indicators such as revenue, product sales, inventory levels, supplier performance, logistics efficiency, manufacturing costs, and quality metrics.

Using **Power BI**, **Power Query**, and **DAX**, the data was cleaned, transformed, and analysed to create six business dashboards: Executive Dashboard, Inventory Analysis Dashboard, Supplier Performance Dashboard, Logistics & Transportation Dashboard, Manufacturing Analysis Dashboard, and Quality Analysis Dashboard.

The insights obtained from this analysis can help management identify trends, monitor performance, reduce operational costs, improve supplier efficiency, optimize inventory management, and enhance overall supply chain effectiveness.

2. PROBLEM STATEMENT

Supply chain operations involve multiple processes such as procurement, inventory management, manufacturing, transportation, supplier management, and quality control. Due to the large volume of operational data, organizations often face challenges in identifying inefficiencies, monitoring performance, and making timely business decisions.

The primary problem addressed in this project is the lack of a centralized analytical system to monitor key supply chain metrics. Without proper visualization and analysis, it becomes difficult to identify high-performing suppliers, optimize inventory levels, reduce transportation costs, monitor manufacturing efficiency, and control product defects.

Therefore, this project aims to analyze the supply chain dataset using Power BI and develop interactive dashboards that provide meaningful insights into business performance. The dashboards enable decision-makers to track important Key Performance Indicators (KPIs), identify trends, and support data-driven decision-making for improving overall supply chain efficiency.

3. OBJECTIVES

The primary objective of this project is to analyze the Supply Chain dataset and develop interactive Power BI dashboards that provide meaningful business insights for effective decision-making.

The specific objectives of this project are:

* To analyze overall supply chain performance using key business metrics.

* To monitor Total Revenue, Products Sold, Manufacturing Cost, Shipping Cost, and Defect Rate.

* To evaluate inventory levels and product availability across different product types.

* To analyze supplier performance based on revenue generation, lead time, shipping cost, and defect rate.

* To study logistics and transportation performance by analyzing shipping time, transportation cost, carriers, and routes.

* To evaluate manufacturing performance through production volume, manufacturing cost, and lead time.

* To analyze product quality using defect rates, inspection results, and transportation modes.

* To create interactive dashboards with slicers and visualizations that enable dynamic business analysis.

* To provide actionable insights and recommendations for improving supply chain efficiency and operational performance.

4. DATASET DESCRIPTION

Dataset Overview

The Supply Chain dataset contains information related to various business operations, including product sales, inventory management, supplier performance, logistics, manufacturing, and quality analysis. The dataset was analyzed using Microsoft Power BI to identify trends, evaluate performance, and generate actionable business insights.

The dataset consists of multiple attributes that help measure the efficiency of the supply chain process from production to product delivery.

Important Features of the Dataset

Some of the major attributes used in this project are:

- Product Type
- SKU (Stock Keeping Unit)
- Price
- Revenue Generated
- Number of Products Sold
- Stock Levels
- Availability
- Order Quantities
- Supplier Name
- Manufacturing Cost
- Manufacturing Lead Time
- Production Volume
- Shipping Cost
- Shipping Time
- Transportation Mode
- Shipping Carrier
- Route
- Defect Rate

- Inspection Results

Purpose of the Dataset

The primary purpose of this dataset is to analyze and monitor the overall performance of the supply chain by evaluating:

- Revenue generation
- Inventory management
- Supplier performance
- Logistics and transportation efficiency
- Manufacturing performance
- Product quality and defect rates

The insights obtained from this analysis help organizations make data-driven decisions, optimize operations, reduce costs, and improve customer satisfaction.

5. TOOLS AND TECHNOLOGIES USED

Microsoft Power BI:

Microsoft Power BI is a powerful Business Intelligence (BI) and Data Visualization tool used to transform raw data into interactive dashboards and meaningful reports. In this project, Power BI was used to create dynamic dashboards that provide insights into supply chain performance.

Power Query:

Power Query is the data transformation and preparation engine in Power BI. It was used to import the dataset, clean inconsistent data, transform columns, and prepare the data for analysis.

DAX (Data Analysis Expression)

DAX is a formula language used in Power BI to create calculated columns and measures. In this project, DAX was used to calculate Key Performance Indicators (KPIs) such as:

- Total Revenue
- Total Products Sold
- Average Shipping Cost
- Average Manufacturing Cost
- Average Defect Rate
- Average Stock Levels
- Production Volume

These measures enabled dynamic and interactive analysis across different dashboards.

MySQL Database:

The project dataset was stored in a MySQL database and connected to Power BI for data retrieval. The database served as the primary data source for analysis and visualization.

Data Visualization Techniques:

Various visualization techniques were used to present the data effectively, including:

- Bar Charts
- Column Charts

- Line Charts
- Donut Charts
- Treemap Charts
- KPI Cards
- Slicers

These visualizations help users quickly understand trends, compare performance, and make informed business decisions.

6. METHODOLOGY

Project Methodology:

The Supply Chain Analysis project was carried out using a systematic data analytics approach to transform raw data into meaningful business insights. The complete workflow followed in this project is described below.

Step 1: Data Collection

The supply chain dataset was collected from the client database. It contains information related to products, suppliers, inventorqy, manufacturing, logistics, transportation, and quality analysis.

Step 2: Data Import

The dataset was connected to Microsoft Power BI and imported into the Power Query Editor for further processing and analysis.

Step 3: Data Cleaning and Preparation

The imported data was cleaned and prepared by performing the following tasks:

- Verified data types for all columns.
- Checked for missing or inconsistent values.
- Renamed columns where necessary for better readability.
- Ensured the dataset was suitable for analysis and visualization.

Step 4: Data Transformation

Power Query was used to transform the data into an analysis-ready format. Required columns were organized and prepared for dashboard development.

Step 5: Data Modelling and DAX Measures

Key Performance Indicators (KPIs) were created using DAX measures, including:

- Total Revenue
- Total Products Sold
- Average Shipping Cost
- Average Manufacturing Cost
- Average Manufacturing Lead Time
- Average Defect Rate
- Average Stock Levels
- Total Production Volume

These measures enabled dynamic calculations and interactive reporting.

Step 6: Dashboard Development

Six interactive dashboards were developed in Microsoft Power BI:

- Executive Supply Chain Dashboard
- Inventory Analysis Dashboard
- Supplier Performance Dashboard
- Logistics & Transportation Dashboard
- Manufacturing Analysis Dashboard
- Quality Analysis Dashboard

Each dashboard includes KPI cards, charts, and slicers for detailed analysis.

Step 7: Business Insights and Recommendations

The dashboards were analyzed to identify trends, evaluate operational performance, and generate business insights. Based on these findings, recommendations were developed to improve inventory management, supplier performance, logistics efficiency, manufacturing operations, and product quality.

7. DASHBOARD ANALYSIS

7.1 Executive Supply Chain Dashboard

Overview:

The Executive Supply Chain Dashboard provides a high-level overview of the organization's overall supply chain performance. It enables management to monitor key business metrics and make strategic decisions based on real-time insights.

Key Performance Indicators (KPIs)

- The dashboard includes the following KPIs:
- Total Revenue
- Total Products Sold
- Average Shipping Cost
- Average Manufacturing Cost
- Average Defect Rate

These KPIs provide a quick summary of the organization's operational performance.

Visualizations Used

The dashboard consists of the following visualizations:

- Revenue by Product Type
- Products Sold by Product Type
- Revenue by Supplier
- Defect Rate by Supplier

Interactive slicers such as **Product Type**, **Supplier Name**, and **Transportation Mode** allow users to filter and analyze the data dynamically.

Business Insights

- Different product types contribute differently to overall revenue.
- Supplier performance has a significant impact on revenue generation.
- Manufacturing and shipping costs influence overall profitability.
- Defect rates should be monitored continuously to maintain product quality.
- Interactive filtering helps management analyze specific suppliers or product categories and make informed business decisions.

Conclusion

The Executive Dashboard serves as a centralized view of the entire supply chain, enabling decision-makers to monitor critical KPIs and identify areas that require operational improvements.

7.2 Inventory Analysis Dashboard

Overview

The Inventory Analysis Dashboard focuses on monitoring inventory performance and product availability across different product categories and suppliers. Effective inventory management helps organizations reduce stock shortages, avoid overstocking, and improve customer satisfaction.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Average Stock Level
- Average Availability
- Total Order Quantity

These KPIs provide a quick overview of inventory status and order demand.

Visualizations Used

The dashboard consists of the following visualizations:

- Stock Levels by Product Type
- Order Quantities by Product Type
- Availability by Product Type
- Stock Levels by Supplier

Interactive slicers such as **Product Type** and **Supplier Name** allow users to filter and analyze inventory information dynamically.

Business Insights

- Inventory levels vary across different product categories.
- Product availability directly impacts customer demand and sales.
- Some suppliers maintain higher stock levels than others.

- Order quantities help identify high-demand products and support inventory planning.
- Proper inventory monitoring can reduce stock-outs and excess inventory costs.

Conclusion

The Inventory Analysis Dashboard enables organizations to optimize inventory management by monitoring stock levels, product availability, and supplier performance. The insights support better planning, efficient resource utilization, and improved operational efficiency.

7.3 Supplier Performance Dashboard

Overview

The Supplier Performance Dashboard evaluates the efficiency and effectiveness of suppliers by analyzing key metrics such as revenue generation, shipping cost, lead time, and defect rate. Monitoring supplier performance helps organizations select reliable suppliers, reduce operational risks, and improve overall supply chain efficiency.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Average Lead Time
- Average Shipping Cost
- Average Defect Rate

These KPIs provide a quick assessment of supplier performance and operational quality.

Visualizations Used

The dashboard consists of the following visualizations:

- Revenue by Supplier
- Shipping Cost by Supplier
- Lead Time by Supplier
- Defect Rate by Supplier

Interactive slicers such as **Supplier Name** enable users to analyze the performance of individual suppliers and compare them effectively.

Business Insights

- Suppliers contribute differently to overall revenue generation.
- Shipping costs vary across suppliers, affecting operational expenses.
- Lead time analysis helps identify suppliers with faster delivery performance.
- Defect rates indicate product quality and supplier reliability.
- Continuous supplier performance monitoring supports better vendor selection and improves supply chain efficiency.

Conclusion

The Supplier Performance Dashboard provides valuable insights into supplier operations and enables management to evaluate supplier reliability, optimize procurement strategies, and improve overall business performance through data-driven decision-making.

7.4 Logistics & Transportation Dashboard

Overview

The Logistics & Transportation Dashboard focuses on analyzing the efficiency of shipping operations and transportation activities within the supply chain. It provides insights into shipping time, shipping cost, transportation cost, carriers, and transportation routes, enabling organizations to optimize logistics performance and reduce operational expenses.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Average Shipping Time
- Average Shipping Cost
- Average Transportation Cost

These KPIs help measure the overall efficiency and cost-effectiveness of logistics operations.

Visualizations Used

The dashboard consists of the following visualizations:

- Shipping Time by Carrier
- Transportation Cost by Mode
- Shipping Cost by Carrier
- Transportation Cost by Route

Interactive slicers such as **Transportation Mode** and **Shipping Carrier** allow users to filter and analyze logistics performance dynamically.

Business Insights

- Different shipping carriers have varying shipping times and costs.
- Transportation modes such as Air, Road, Rail, and Sea contribute differently to overall transportation expenses.
- Route analysis helps identify cost-intensive transportation paths.
- Efficient logistics planning can reduce delivery time and operational costs.
- Monitoring logistics performance supports better resource allocation and customer satisfaction.

Conclusion

The Logistics & Transportation Dashboard enables organizations to evaluate shipping performance, optimize transportation strategies, reduce logistics costs, and improve overall supply chain efficiency through data-driven decision-making.

7.5 Manufacturing Analysis Dashboard

Overview

The Manufacturing Analysis Dashboard focuses on evaluating the efficiency of the manufacturing process by analyzing production volume, manufacturing cost, and manufacturing lead time. It helps organizations monitor production performance, optimize manufacturing operations, and reduce operational costs.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Average Manufacturing Cost
- Average Manufacturing Lead Time
- Total Production Volume

These KPIs provide a summary of the overall manufacturing performance and operational efficiency.

Visualizations Used

The dashboard consists of the following visualizations:

- Production Volume by Product Type
- Manufacturing Lead Time by Product Type
- Manufacturing Cost by Product Type
- Production Volume by Supplier

Interactive slicers such as **Product Type** and **Supplier Name** allow users to filter and analyze manufacturing performance dynamically.

Business Insights

- Production volumes vary across different product categories.
- Manufacturing costs differ based on product type and production requirements.
- Manufacturing lead time impacts the overall supply chain efficiency and delivery schedules.
- Supplier performance influences production capacity and output.
- Monitoring manufacturing metrics helps improve productivity, reduce costs, and optimize resource utilization

Conclusion

The Manufacturing Analysis Dashboard provides valuable insights into production efficiency and manufacturing operations. It enables management to monitor costs, evaluate production performance, and make data-driven decisions to improve operational effectiveness and support business growth.

7.6 Quality Analysis Dashboard

Overview

The Quality Analysis Dashboard focuses on evaluating product quality and inspection performance across the supply chain. It helps organizations monitor defect rates, inspection results, and transportation-related quality issues to ensure high product standards and customer satisfaction.

Key Performance Indicators (KPIs)

The dashboard includes the following KPIs:

- Average Defect Rate
- Total Inspection Results
- Total Products Sold

These KPIs provide an overview of product quality and the effectiveness of the inspection process.

Visualizations Used

The dashboard consists of the following visualizations:

- Defect Rate by Product Type
- Inspection Results Distribution
- Defect Rate by Supplier
- Defect Rate by Transportation Mode

Interactive slicers such as **Product Type** and **Supplier Name** allow users to analyze quality performance for different products and suppliers.

Business Insights

- Defect rates vary across different product categories.
- Inspection results help identify products that require additional quality control.
- Supplier-wise defect analysis assists in evaluating supplier reliability.
- Transportation modes may influence product quality and defect occurrence.
- Continuous monitoring of quality metrics helps reduce defects, improve customer satisfaction, and strengthen supply chain performance.

Conclusion

The Quality Analysis Dashboard provides a comprehensive view of product quality and inspection performance. It enables management to identify quality issues, evaluate supplier performance, and implement corrective actions to improve overall supply chain efficiency and product reliability.

8. KEY BUSINESS INSIGHTS

The analysis of the Supply Chain dataset provided several valuable business insights that can support strategic and operational decision-making.

Revenue and Sales Performance

- Different product categories contribute differently to total revenue generation.
- Product sales performance varies across categories, indicating differences in customer demand.
- Revenue generated by suppliers highlights the importance of supplier relationships in overall business performance.

Inventory Management Insights

- Stock levels vary across product types and suppliers.
- Certain products maintain higher availability than others, impacting customer satisfaction and sales.
- Effective inventory monitoring helps reduce stock shortages and excess inventory costs.

Supplier Performance Insights

- Supplier performance differs in terms of revenue contribution, shipping cost, lead time, and defect rates.
- Some suppliers demonstrate better reliability and operational efficiency.
- Monitoring supplier performance enables better procurement decisions and supplier selection.

Logistics and Transportation Insights

- Transportation costs vary based on transportation modes and routes.
- Shipping carriers differ in delivery performance and shipping costs.

- Optimizing transportation routes and carrier selection can reduce logistics expenses and improve delivery efficiency.

Manufacturing Insights

- Manufacturing costs vary across product categories.
- Production volumes differ among suppliers and product types.
- Manufacturing lead time directly impacts supply chain responsiveness and customer satisfaction.

Quality Management Insights

- Defect rates vary across product types, suppliers, and transportation modes.
- Inspection results indicate the effectiveness of quality control processes.
- Continuous monitoring of quality metrics can help reduce product defects and improve customer satisfaction.

Overall Business Impact

The integrated analysis of inventory, supplier performance, logistics, manufacturing, and quality metrics provides a comprehensive understanding of supply chain operations. The dashboards enable data-driven decision-making and support continuous improvement across the organization.

9. RECOMMENDATIONS

Based on the analysis of the Supply Chain dataset and the insights obtained from the dashboards, the following recommendations are proposed to improve overall supply chain performance:

Optimize Inventory Management

- Continuously monitor stock levels to prevent stock shortages and overstock situations.
- Use demand forecasting techniques to maintain optimal inventory levels.
- Improve inventory planning based on product demand and sales trends.

Improve Supplier Performance

- Regularly evaluate suppliers based on revenue contribution, lead time, shipping cost, and defect rates.

- Develop long-term relationships with high-performing suppliers.
- Provide performance feedback and corrective actions for underperforming suppliers.

Reduce Logistics and Transportation Costs

- Optimize transportation routes to minimize delivery costs.
- Select shipping carriers based on cost and delivery performance.
- Analyze transportation modes to identify the most cost-effective options.

Enhance Manufacturing Efficiency

- Monitor manufacturing lead time and production volume to improve operational efficiency.
- Identify opportunities to reduce manufacturing costs without affecting product quality.
- Improve production planning to meet customer demand effectively.

Strengthen Quality Control

- Increase the frequency of product inspections to identify defects at an early stage.
- Monitor defect rates across suppliers and product categories.
- Implement preventive quality assurance measures to reduce defective products.

Adopt Data-Driven Decision Making

- Utilize interactive Power BI dashboards for continuous performance monitoring.
- Track Key Performance Indicators (KPIs) regularly to support strategic planning.
- Encourage management to make decisions based on analytical insights rather than assumptions.

Overall Recommendation

Implementing these recommendations will improve supply chain efficiency, reduce operational costs, enhance product quality, strengthen supplier relationships, and increase customer satisfaction. Data-driven decision-making through Power BI dashboards will enable the organization to achieve better operational and business outcomes.

10. CONCLUSION

The “**Supply Chain Analysis Using Power BI**” project successfully transformed raw supply chain data into meaningful business insights through interactive dashboards and analytical reports. The project provided a comprehensive analysis of various aspects of the supply chain, including revenue, inventory management, supplier performance, logistics and transportation, manufacturing operations, and product quality.

By utilizing Microsoft Power BI, Power Query, and DAX, the data was effectively cleaned, transformed, and visualized into six interactive dashboards. These dashboards enable decision-makers to monitor Key Performance Indicators (KPIs), identify trends, compare performance across different product categories and suppliers, and make informed business decisions.

The analysis highlighted opportunities for improving inventory management, optimizing supplier performance, reducing logistics costs, enhancing manufacturing efficiency, and strengthening quality control processes. The recommendations provided in this report can help organizations improve operational efficiency, reduce costs, and increase customer satisfaction.

Overall, this project demonstrates the practical application of Business Intelligence and Data Analytics techniques in solving real-world supply chain problems. It showcases how data-driven decision-making can support strategic planning and contribute to improved business performance and sustainable growth.

Figure 1: Executive Supply Chain Dashboard

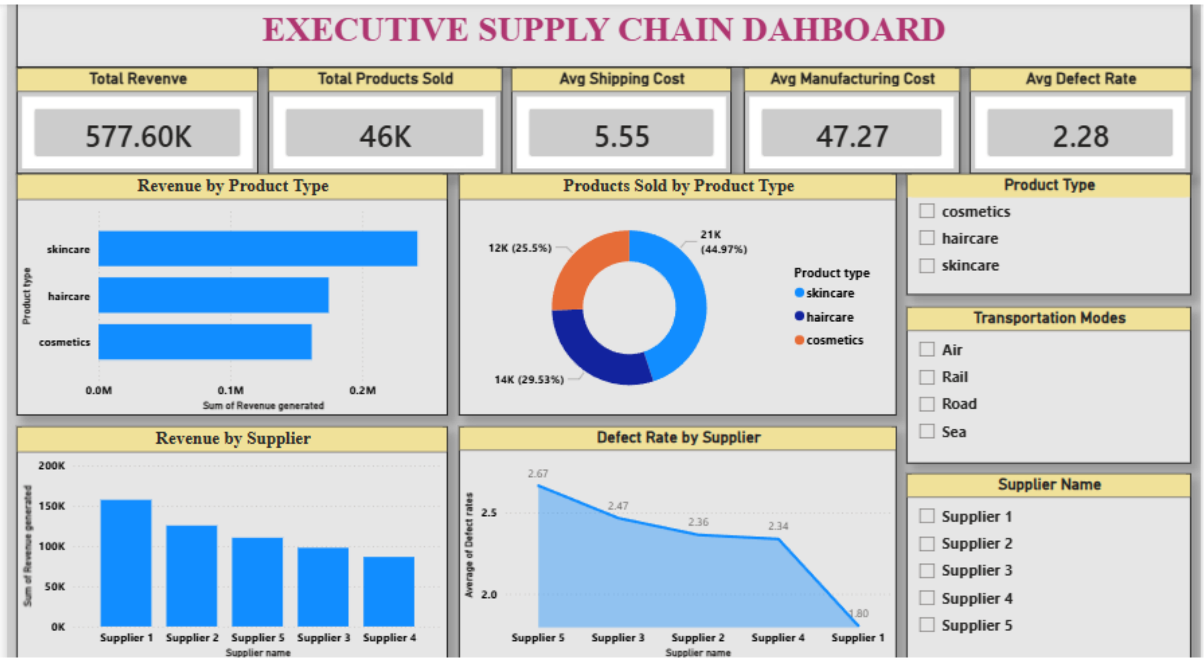

Figure 2: Inventory Analysis Dashboard

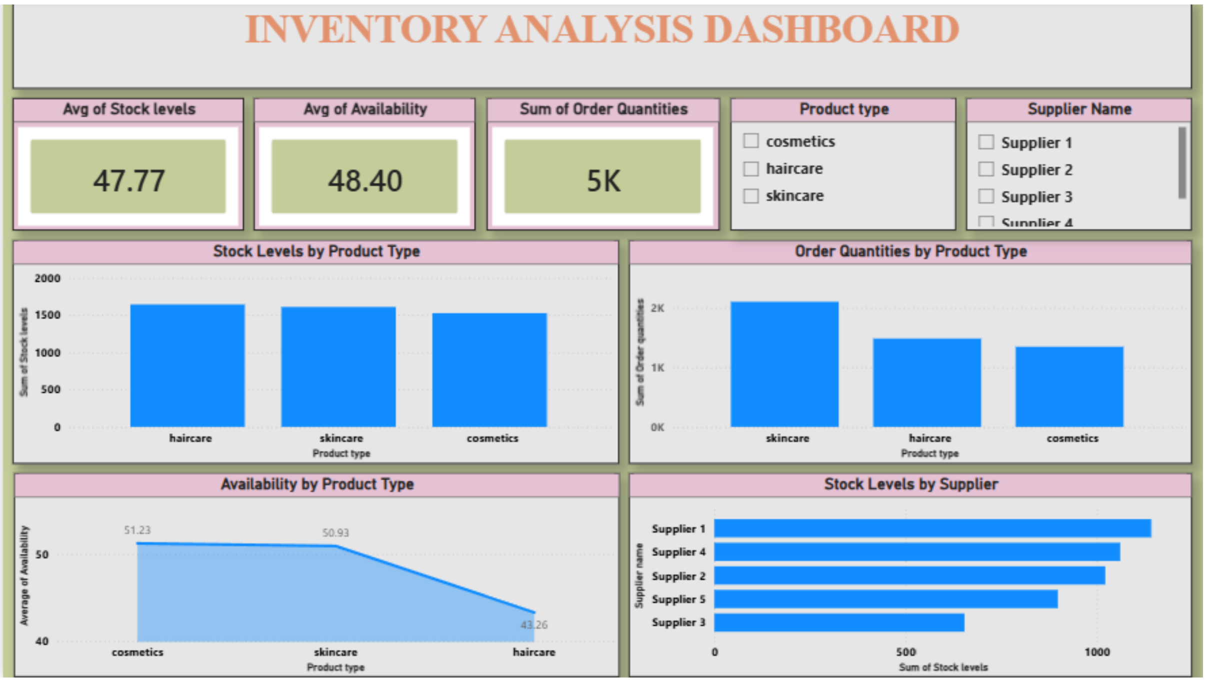

Figure 3: Supplier Performance Dashboard

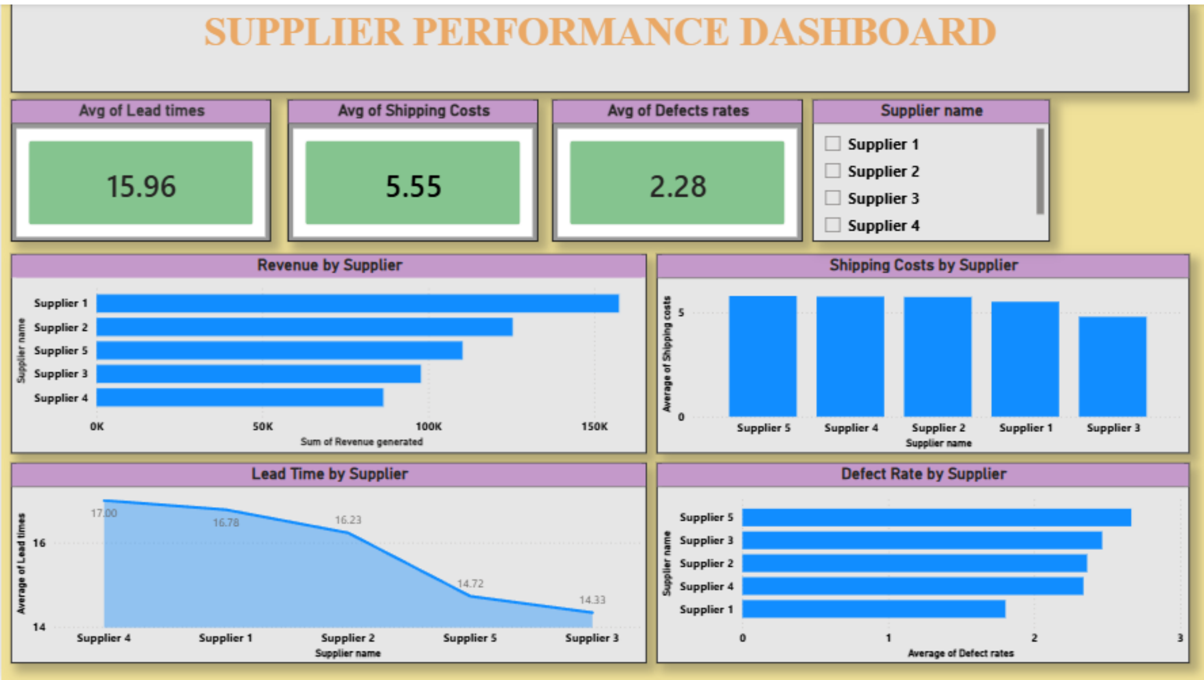

Figure 4: Logistics & Transportation Dashboard

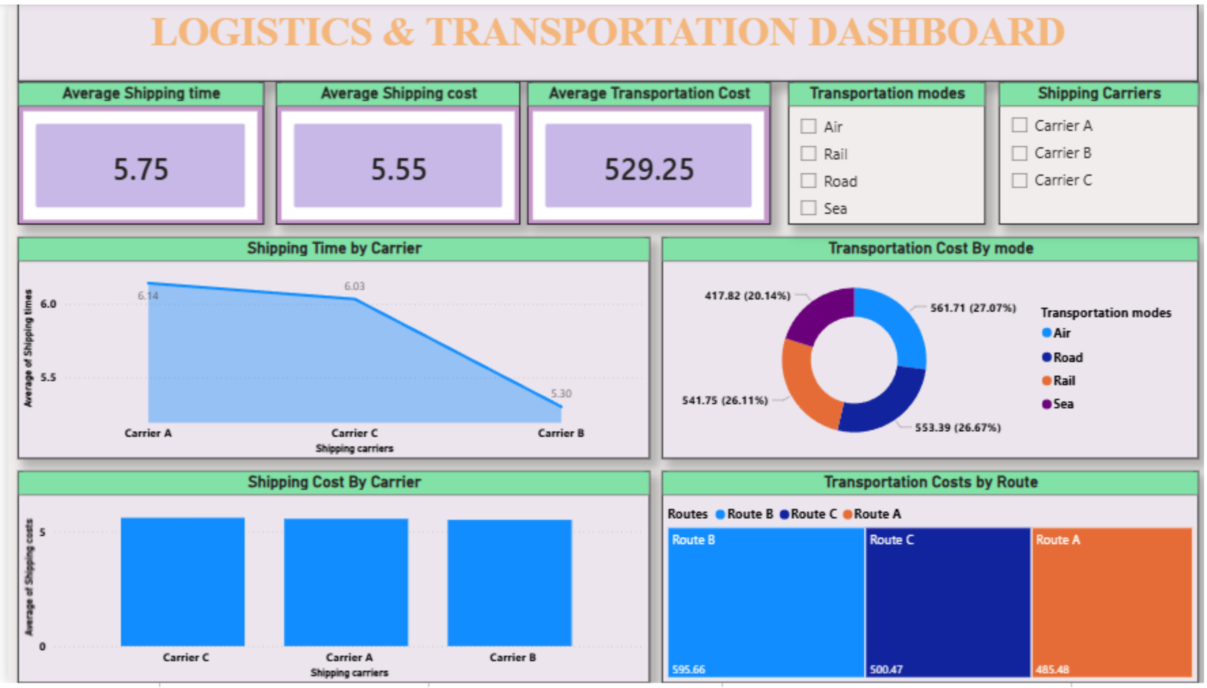

Figure 5: Manufacturing Analysis Dashboard

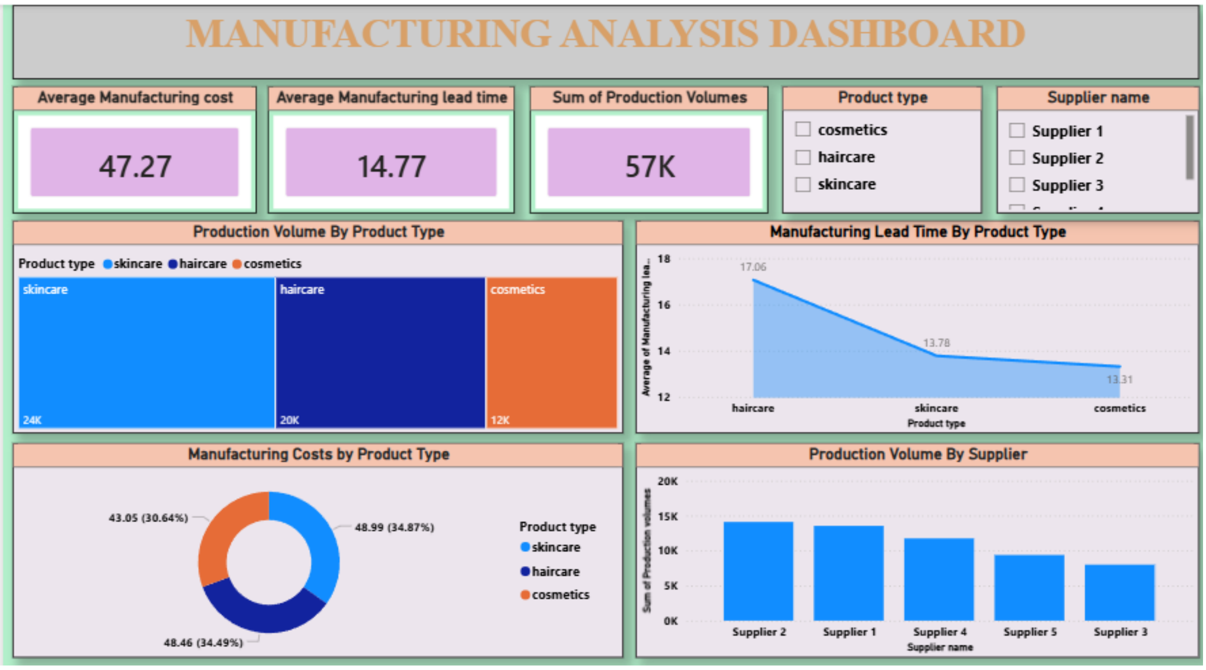

Figure 6: Quality Analysis Dashboard